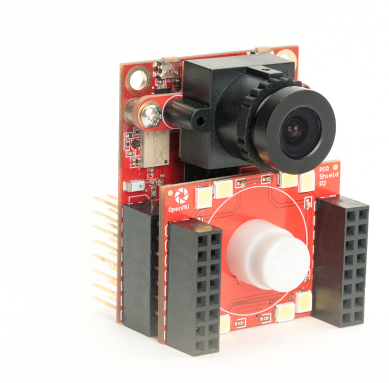
OpenMV® PIR Shield

Datasheet

SKU: OMV-SHIELD-PIR

Description

The PIR Shield gives your OpenMV Cam the ability to wake up from deep sleep on movement by pulling P10 low. It features eight PWM-dimmable high-power white LEDs (controlled by P7) and eight PWM-dimmable high-power IR LEDs at 850 nm (controlled by P8). The PIR sensor was specially chosen for its ultra-low current consumption of 6 \u00B5A (compared to >100 \u00B5A for most PIR sensors), enabling years of battery operation in deep sleep with cameras like the RT1062.



Target Areas

Motion-triggered capture, security cameras, wildlife monitoring, battery-powered surveillance, deep-sleep wakeup applications.

Key Features

- **Ultra-low-power PIR** — 6 µA PIR sensor — enables years of battery life in deep sleep.
- **Deep sleep wakeup** — Wakes OpenMV Cam from deep sleep on motion by pulling P10 low.
- **8 white LEDs** — PWM-dimmable high-power white LEDs on P7.
- **8 IR LEDs** — PWM-dimmable high-power 850 nm IR LEDs on P8.
- **Flexible wakeup pin** — PIR wakeup pin changeable to P9 via solder bridge jumpers.
- **Works without deep sleep** — PIR signal usable as standard GPIO interrupt on cameras without low-power wakeup.

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1 Features

1.1 PIR Motion Sensor

Component	Details
Current Consumption	6 µA — ultra-low-power, suitable for multi-year battery operation.
Wakeup	Pulls P10 low on motion detection. Wakes camera from deep sleep. Pin changeable to P9 via solder bridge.
Power	For non-dual-row OpenMV Cams, connect the RAW power input pin on the module to 3.3 V to power the PIR sensor.

1.2 LED Illumination

Component	Details
White LEDs	8 high-power white LEDs. PWM-dimmable via P7. Disconnectable.
IR LEDs	8 high-power 850 nm IR LEDs. PWM-dimmable via P8. Disconnectable.

1.3 Compatibility

Component	Details
Supported Cameras	All OpenMV Cam models. Optimized for cameras with deep sleep capability (e.g., OpenMV Cam RT1062). Also works as GPIO interrupt on cameras without low-power wakeup.

2 General Info

Dimension	Value
Weight	8 g

3 Company Information

Field	Details
Website	https://openmv.io

4 Reference Documentation

Reference	Link / Document
Product Page	https://openmv.io/products/pir-shield
Documentation	https://docs.openmv.io

5 Revision History

Date	Revision	Changes
2026-04-04	Rev 1	Initial public release.